Primary Gaussian Phases and All-Degree Frobenius Dynamics for the Elliptic Curve $y^2 = x^3 - x$

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Abstract

For the reductions of $y^2 = x^3 - x$ at every odd prime, we give a sign-complete derivation of the Frobenius eigenvalues. At split primes, the centralizer of an intrinsic order-four automorphism places Frobenius in the Gaussian order, and an explicit rational point of order four fixes the primary Gaussian unit. At inert primes a character involution gives trace zero. The resulting unordered eigenvalue pair is determined without a fitted phase or a point-count sign convention. We then prove irrational normalized phase in the ordinary chamber, exact phase order four in the supersingular chamber, and classify every extension degree attaining a Hasse endpoint. All extension-field counts and primitive closed-point counts follow, with a complete repetition dictionary.

Keywords: complex multiplication; Frobenius dynamics; Gaussian integers; elliptic curves; primary normalization; closed points

中文摘要

本文统一处理曲线 $y^2 = x^3 - x$ 在所有奇素数处的约化。分裂情形中，四阶自同态的中心化子迫使弗罗贝尼乌斯位于高斯整数环；显式有理四阶点使群阶被八整除，从而唯一确定主高斯整数的实部符号。惯性情形通过二次特征的反号对合证明迹为零，整个相位约定无需拟合。普通约化的归一化相位与圆周无理旋转对应，超奇异约化的相位恰有四阶。由此分类全部扩域次数的哈塞端点，并用轨道分解与莫比乌斯反演给出每个次数的闭点计数，明确区分本原轨道与重复。
关键词：复乘；弗罗贝尼乌斯动力学；高斯整数；椭圆曲线；主元归一化；闭点计数

1 Frozen object, ownership, and the first advance

Round 0 advance: an all-odd-prime trace formula with a proved Gaussian-unit convention. Let p be an odd prime, let E_p denote the smooth projective curve $y^2 = x^3 - x$ over $\mathbf{F}_p$, and let

$$F : E_p(\overline{\mathbf{F}}_p) \longrightarrow E_p(\overline{\mathbf{F}}_p), \qquad F(x, y) = (x^p, y^p).$$

This formula fixes the Frobenius convention; its cohomological pullback has eigenvalue p on degree two. We put $N_{p,n} = \#E_p(\mathbf{F}_{p^n})$ and $t_p = p + 1 - N_{p,1}$. The prime 2 is excluded: the integral model has discriminant 64. The parameter p indexes reductions of one curve and is not an autonomous real-time coordinate.

The general closed-point/zeta dictionary was already owned locally by flow papers 4 and 6 for a projective-line Frobenius suspension. The earlier C41 cubic-CM package considered $y^2 = x^3 + 1$, an inert-prime trace law and finite extension checks. The present owner is the quartic-CM all-prime primary phase, its ordinary/supersingular distinction, and its complete degree consequences. These results specialize classical elliptic-curve theory; they are not presented as newly discovered CM theorems. We use the endomorphism and finite-field results in Milne [1], with every special sign step proved below.

Theorem 1 (Sign-complete CM trace). *If $p \equiv 3 \pmod 4$, then $t_p = 0$. If $p \equiv 1 \pmod 4$, there are unique integers a, b satisfying*

$$b > 0, \qquad a \text{ odd}, \qquad b \text{ even}, \qquad a^2 + b^2 = p, \qquad a + b \equiv 1 \pmod 4),$$

and $t_p = 2a$. The unordered Frobenius eigenvalues are $a \pm bi$ in the split case and $\pm i\sqrt{p}$ in the inert case.

Proof. Let χ be the quadratic character extended by $\chi(0) = 0$. Counting the two square roots of a nonzero square gives

$$N_{p,1} = p + 1 + \sum_{x \in \mathbf{F}_p} \chi(x^3 - x).$$

When $p \equiv 3 \pmod 4$, replacing x by $-x$ negates the sum, so it vanishes as an integer. This includes the good prime $p = 3$.

For $p \equiv 1 \pmod 4$, choose $j \in \mathbf{F}_p$ with $j^2 = -1$. The endomorphism $I(x, y) = (-x, jy)$ satisfies $I^2 = [-1]$ and commutes with F. By the elliptic endomorphism classification, the endomorphism algebra containing $\mathbf{Q}(I)$ is either that quadratic field or a quaternion algebra. In either case the centralizer of I is $\mathbf{Q}(I)$: in the latter case write the algebra as $\mathbf{Q}(i) \oplus \mathbf{Q}(i)v$ with $vi = -iv$ and compare the two coefficients. Thus F lies in $\mathbf{Q}(i)$. Its integrality places it in the maximal order $\mathbf{Z}[i]$, so $F = a + bi$, and degree and trace give $a^2 + b^2 = p$ and $t_p = 2a$.

All three nonzero 2-torsion points are rational. Therefore $F - [1]$ kills $E[2]$ and factors through the separable isogeny [2]. The endomorphism $(F - [1])/2$ is again an integral element of $\mathbf{Q}(i)$. It follows that a is odd and b even.

Now $Q = (j, j - 1)$ lies on the curve because $(j - 1)^2 = -2j = j^3 - j$. The tangent slope is

$$m = \frac{3j^2 - 1}{2(j - 1)} = j + 1, \qquad x(2Q) = m^2 - 2j = 0, \qquad y(2Q) = mj - (j - 1) = 0.$$

Hence $2Q = (0, 0)$, and Q has order four. The independent point $(1, 0)$ of order two gives an eight-element subgroup, so $8 \mid p+1-2a$. If $b \equiv 0 \pmod 4$ then $p \equiv 1 \pmod 8$ and $a \equiv 1 \pmod 4$; if $b \equiv 2 \pmod 4$ then $p \equiv 5 \pmod 8$ and $a \equiv 3 \pmod 4$. These are precisely $a + b \equiv 1 \pmod 4$. Unique factorization in $\mathbf{Z}[i]$ fixes $|a|, |b|$ and this congruence fixes the sign of a. Taking $b > 0$ chooses the upper display root. It does not select one literal prime ideal: both conjugates satisfy the primary congruence modulo $(1 + i)^3$. □

p	a	b	t_p	chamber
3	—	—	0	supersingular
5	−1	2	−2	ordinary
13	3	2	6	ordinary
17	1	4	2	ordinary
29	−5	2	−10	ordinary

The negative real part at $p = 5$ is essential: choosing the positive representation $1 + 2i$ changes the curve trace. Complex conjugation, however, changes neither trace nor the unordered eigenvalue pair.

2 Phase rigidity, extension counts, and primitive orbits

Round 1 advance: phase periodicity is classified, then all degrees of fixed and primitive points are determined. Let α_p be the eigenvalue with positive imaginary part and let $\theta_p \in (0, \pi)$ satisfy $\alpha_p/\sqrt{p} = e^{i\theta_p}$.

Theorem 2 (Phase and Hasse endpoints)**.** *For $p \equiv 1 \pmod 4$ the curve is ordinary and θ_p/π is irrational. For $p \equiv 3 \pmod 4$ it is supersingular and its normalized eigenvalues have exact order four. The Hasse bound over $\mathbf{F}_{p^n}$ is strict for every n in the ordinary chamber; it is saturated exactly for even n in the supersingular chamber. At $n = 2m$ in that chamber,*

$$N_{p,2m} = \left(p^m - (-1)^m\right)^2.$$

Proof. The standard ordinary criterion is $p \nmid t_p$ for a curve over $\mathbf{F}_p$ [2]. In the split chamber, $0 < |a| < \sqrt{p}$ gives $p \nmid 2a$; in the inert chamber the trace is zero. If $\alpha_p/\sqrt{p}$ were a root of unity in the split chamber, its square $\alpha_p/\overline{\alpha_p}$ would be a root of unity in $\mathbf{Q}(i)$, hence $1, -1, i$, or $-i$. Those cases respectively force $b = 0$, $a = 0$, $a = b$, or $a = -b$, all incompatible with an odd prime norm and the parity just established. Thus the phase is irrational. The trace over degree n is $2p^{n/2}\cos(n\theta_p)$, so equality at a Hasse endpoint would make $n\theta_p$ an integer multiple of π, which is impossible. In the inert chamber the eigenvalues are $\pm i\sqrt{p}$: the trace is zero for odd n and $2(-p)^{n/2}$ for even n. Substitution yields the displayed square. □

Theorem 3 (All-degree ledger)**.** *Put $S_{p,0} = 2$, $S_{p,1} = t_p$, and $S_{p,n} = t_p S_{p,n-1} - pS_{p,n-2}$. For every integer $n \geq 1$,*

$$N_{p,n} = p^n + 1 - S_{p,n}, \qquad B_{p,n} = \frac{1}{n}\sum_{d \mid n} \mu(d) N_{p,n/d} \in \mathbf{Z}_{\geq 0},$$

where $B_{p,n}$ counts the primitive F-orbits of length n.

Proof. The Frobenius trace theorem [1, IV, Theorem 9.10 and Remark 9.11] gives $N_{p,n} = 1 + p^n - \alpha_p^n - \overline{\alpha_p}^n$. The quadratic characteristic polynomial gives the recurrence. Every geometric point has coordinates in a finite extension of $\mathbf{F}_p$, and hence lies on a finite Frobenius orbit. Such orbits are exactly the closed points. A length-d orbit contributes d fixed points to F^n if and only if $d \mid n$, giving $N_{p,n} = \sum_{d|n} dB_{p,d}$. Möbius inversion proves the formula, and the orbit interpretation proves integrality and nonnegativity for every degree without a finite cutoff. □

The orbit convention uses forward F, primitive weight one and phase zero; F^n records repetitions. The cohomological CM phase is not a complex weight attached to each closed point. The differential of the p-power morphism is zero, which must not be interpreted as the stability of a real differentiable flow. Completeness belongs to the closed-point theorem; numerical tests cannot replace it. Since $F^n - [1]$ has differential $-\mathrm{id}$, it is separable and its kernel is reduced. Thus every geometric fixed point has scheme multiplicity one, independently of the real-flow stability question.

3 Conclusion

Rational four-torsion fixes the primary Gaussian sign uniformly at every split odd prime, while a character involution settles every inert prime. This proof identifies the exact source mechanism behind the phase convention. The resulting distinction between irrational ordinary phase and order-four supersingular phase classifies all extension Hasse endpoints and closes every degree of the primitive closed-point ledger.

References

[1] J. S. Milne, *Elliptic Curves*, 2nd ed., World Scientific, 2020. II §6 (endomorphisms), III Example 3.18 (quartic CM), IV §9, Theorems 9.4 and 9.10, Remark 9.11 and Exercise 9.12 (finite-field counts, zeta and functional equation). `https://www.jmilne.org/math/Books/EC2.pdf`.

[2] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd ed., Springer, 2009, V §3 (ordinary and supersingular reduction). `https://doi.org/10.1007/978-0-387-09494-6`.